

SECTION AAnswer **all** questions.

1. (a) Radon is a radioactive gas and emits alpha particles.

(i) Explain what is meant by the *activity* of a radioactive source **and** give its unit. [2]

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(ii) The half-life of radon is 3.3×10^5 s. Determine the time it takes for a sample of radon to decay to 20 % of its initial activity. [3]

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(iii) Discuss the extent to which alpha particles are a risk to the human body. [3]

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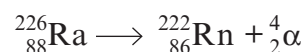
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(b) Radium-226 decays into radon.



Assuming that 98% of the energy released in this process is converted to the kinetic energy of the alpha particle, use the data below to determine the speed of the alpha particle. [5]

Atomic masses: ${}^{226}_{88}\text{Ra} = 226.025410\text{ u}$ ${}^{222}_{86}\text{Rn} = 222.017578\text{ u}$

$${}^4_2\text{He} = 4.002603\text{u} \quad m_{\text{electron}} = 0.000549\text{u}$$
$$1\text{ u} = 931\text{ MeV}$$
